Rafael Mouallem Rosa

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Current Research

My current research develops the Sources of Uncertainty framework for Bayesian inference and explores its applications, with a particular interest in macroeconomics and finance.

Academic Employment

Inria, Statify team — Grenoble, France
Researcher — 2026–present
Postdoctoral Researcher — 2024–2026

Education

Ph.D. in Economics, São Paulo School of Economics, Fundação Getulio Vargas — 2015–2019
Advisor: Victor Filipe Martins-da-Rocha

Visiting Scholar, Harvard University, Department of Economics — 2018–2019
Host: Tomasz Strzalecki

M.S. in Economics, São Paulo School of Economics, Fundação Getulio Vargas — 2013–2015

B.S. in Economics, PUC-SP — 2005–2010

Publications and Manuscripts

Statistics and Machine Learning

- **Rosa, R.M.**, Arbel, J., and Nguyen, H.D. (2026). *Empowering Bayes with sources of uncertainty: from confidence modelling to sparse estimation.* arXiv:2605.03134.
- **Rosa, R.M.**, Arbel, J., and Nguyen, H.D. (2026). *Bayesian inference with sources of uncertainty: a confidence-weighted approach to sparsity.* Accepted workshop paper, OPTIMAL: Optimisation and Post-Bayesian Inference in Machine Learning, AISTATS 2026 Workshop.
- **Rosa, R.M.** and Arbel, J. (2026). *SoU-RHS: Bayesian Structured Sparsity for Compact Architecture Recovery in CNNs.* Manuscript.

Economics and Decision Theory

- **Rosa, R.M.** (2026). *SoU-BVARs: Adapting Inference about Macroeconomic Relationships.* Manuscript.
- Martins-da-Rocha, V.F. and **Rosa, R.M.** (2023). *Complete markets with bankruptcy risk and pecuniary default punishments.* *Economic Theory*, 75(3), 625–640.
- Martins-da-Rocha, V.F. and **Rosa, R.M.** (2020). *Second-Order Beliefs and Second-Order Expected Utility.* HAL preprint.
- Teles, V.K., Júnior, C.J., and **Rosa, R.M.** (2015). *Investment-specific technological change and the Brazilian macroeconomy.* *Brazilian Review of Econometrics*, 35(2), 233–264.

Industry Experience

Safra Asset Management São Paulo, Brazil
Quantitative Portfolio Manager 2020–2022

- Designed and implemented quantitative trading strategies in futures and equities.
- Built research-to-production workflows for signal construction, backtesting, and risk monitoring.

Macro Capital Asset Management São Paulo, Brazil
Partner and Economist 2019–2020

- Developed macro-finance quantitative models for investment decisions and scenario analysis.

Brasil Capital Asset Management São Paulo, Brazil
Analyst 2010–2011

Votorantim Bank São Paulo, Brazil
Intern 2009–2010

Teaching

São Paulo School of Economics, Fundação Getulio Vargas São Paulo, Brazil
Instructor, Probability, undergraduate course 2016

Conference Presentations

- *Empowering Bayes with Sources of Uncertainty: From Confidence Modelling to Structured Sparsity in Neural Networks.* ISBA 2026 World Meeting, Bayesian Deep Learning session, Nagoya, Japan. *Oral presentation.*
- *Empowering Bayes with Sources of Uncertainty.* The 4th Bayes-Duality Workshop, Tokyo, Japan. *Poster presentation.*

Professional Service

Referee: *Bayesian Analysis.*

Awards and Scholarships

- CAPES PDSE Scholarship, Doctorate Abroad, Harvard University 2018–2019
- CAPES Scholarship, Ph.D. in Economics, FGV 2015–2019
- VII INFI–FEBRABAN Award, 3rd place, Category A 2015
- CAPES Scholarship, M.S. in Economics, FGV 2013–2014

Technical Skills

- Python, R, MATLAB, LaTeX